

DW_fp_mac

Floating-Point Multiply-and-Add

Version, STAR, and myDesignWare Subscriptions: [IP Directory](#)

Features and Benefits

- The precision format is parameterizable for either IEEE single, double precision, or a user-defined custom format
- Hardware for denormal numbers of IEEE 754 standard is selectively provided
- Accuracy conforms to IEEE 754 Floating-point standard

Revision History

Description

DW_fp_mac is a floating-point component that performs the multiply and add operation. It sums up a floating-point product of input a and b to input c ($ab + c$) to produce a floating-point multiply and add result, z.

The output of this component has accuracy consistent with the IEEE Standard 754, where the computation happens as it was done using infinite precision, and rounding is executed as a last step to obtain the final result. As a consequence, the accuracy of DW_fp_mac is much better than the accuracy of an implementation using DW_fp_mult and DW_fp_add.

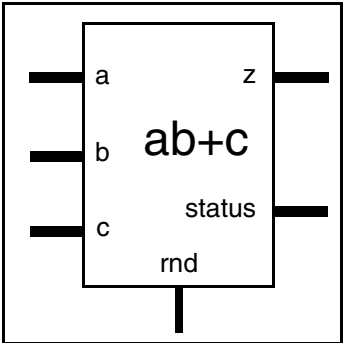


Table 1-1 Pin Description

Pin Name	Width	Direction	Function
a	<i>exp_width</i> + <i>sig_width</i> + 1 bits	Input	Multiplier
b	<i>exp_width</i> + <i>sig_width</i> + 1 bits	Input	Multiplicand
c	<i>exp_width</i> + <i>sig_width</i> + 1 bits	Input	Addend
rnd	3 bits	Input	Rounding mode; supports all rounding modes described in the Datapath Floating-Point Overview
z	<i>exp_width</i> + <i>sig_width</i> + 1 bits	Output	MAC result ($a \times b + c$)
status	8 bits	Output	Status flags for the result z For details, see STATUS Flags in the <i>Datapath Floating-Point Overview</i> .

Table 1-2 Parameter Description

Parameter	Values	Description
sig_width	2 to 253 bits Default: 23	Word length of fraction field of floating-point numbers a, b, c, and z
exp_width	3 to 31 bits Default: 8	Word length of biased exponent of floating-point numbers a, b, c and z
ieee_compliance	0 or 1 Default: 0	Level of support for IEEE 754: 0: No support for IEEE 754 NaNs and denormals; NaNs are considered infinities and denormals are considered zeros 1: Fully compliant with the IEEE 754 standard, including support for NaNs and denormals For more, see IEEE 754 Compatibility in the <i>Datapath Floating-Point Overview</i>.

Table 1-3 Synthesis Implementations

Implementation Name	Function	License Feature Required
rtl	Area-optimized synthesis model	DesignWare
str	Delay-optimized synthesis model	DesignWare

Table 1-4 Simulation Models

Model	Function
DW02.DW_FP_MAC_CFG_SIM	Design unit name for VHDL simulation
dw/dw02/src/DW_fp_mac_sim.vhd	VHDL simulation model source code
dw/sim_ver/DW_fp_mac.v	Verilog simulation model source code

Suppressing Warning Messages During Verilog Simulation

The Verilog simulation model includes macros that allow you to suppress warning messages during simulation.

To suppress all warning messages for all DWBB components, define the DW_SUPPRESS_WARN macro in either of the following ways:

- Specify the Verilog preprocessing macro in Verilog code:

```
`define DW_SUPPRESS_WARN
```

- Or, include a command line option to the simulator, such as:

```
+define+DW_SUPPRESS_WARN (which is used for the Synopsys VCS simulator)
```

The warning messages for this model include the following:

- If an invalid rounding mode has been detected on `rnd`, the following message is displayed:

```
WARNING: <instance_path>:  
        at time = <timestamp>: Illegal rounding mode.
```

To suppress this message, use the DW_SUPPRESS_WARN macro explained earlier.

Related Topics

- [Datapath Floating-Point Overview](#)
- [DesignWare Building Block IP User Guide](#)

HDL Usage Through Component Instantiation - VHDL

```
library IEEE,DWARE;
use IEEE.std_logic_1164.all;
use DWARE.DWpackages.all;
use DWARE.DW_Foundation_comp_arith.all;

entity DW_fp_mac_inst is
    generic (
        inst_sig_width : POSITIVE := 23;
        inst_exp_width  : POSITIVE := 8;
        inst_ieee_compliance : INTEGER := 0
    );
    port (
        inst_a : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_b : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_c : in std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        inst_rnd : in std_logic_vector(2 downto 0);
        z_inst : out std_logic_vector(inst_sig_width+inst_exp_width downto 0);
        status_inst : out std_logic_vector(7 downto 0)
    );
end DW_fp_mac_inst;

architecture inst of DW_fp_mac_inst is
begin

    -- Instance of DW_fp_mac
    U1 : DW_fp_mac
    generic map (
        sig_width => inst_sig_width,
        exp_width => inst_exp_width,
        ieee_compliance => inst_ieee_compliance
    )
    port map (
        a => inst_a,
        b => inst_b,
        c => inst_c,
        rnd => inst_rnd,
        z => z_inst,
        status => status_inst
    );

end inst;

-- pragma translate_off
configuration DW_fp_mac_inst_cfg_inst of DW_fp_mac_inst is
for inst
end for; -- inst
end DW_fp_mac_inst_cfg_inst;
```

```
-- pragma translate_on
```

HDL Usage Through Component Instantiation - Verilog

```
module DW_fp_mac_inst( inst_a, inst_b, inst_c, inst_rnd, z_inst, status_inst );

parameter inst_sig_width = 23;
parameter inst_exp_width = 8;
parameter inst_ieee_compliance = 0;

input [inst_sig_width+inst_exp_width : 0] inst_a;
input [inst_sig_width+inst_exp_width : 0] inst_b;
input [inst_sig_width+inst_exp_width : 0] inst_c;
input [2 : 0] inst_rnd;
output [inst_sig_width+inst_exp_width : 0] z_inst;
output [7 : 0] status_inst;

// Instance of DW_fp_mac
DW_fp_mac #(inst_sig_width, inst_exp_width, inst_ieee_compliance) U1 (
    .a(inst_a),
    .b(inst_b),
    .c(inst_c),
    .rnd(inst_rnd),
    .z(z_inst),
    .status(status_inst) );

endmodule
```

Revision History

For notes about this release, see the [DesignWare Building Block IP Release Notes](#).

For lists of both known and fixed issues for this component, refer to the [STAR report](#).

For a version of this datasheet with visible change bars, click [here](#).

Date	Release	Updates
September 2024	DWBB_202409.0	■ Updated date and version; no technical changes
December 2023	DWBB_202312.0	■ Updated date and version; no technical changes
July 2020	DWBB_201912.5	■ Adjusted the description of the <i>ieee_compliance</i> parameter in Table 1-2 on page 2 ■ Added “Suppressing Warning Messages During Verilog Simulation” on page 3 ■ Added this Revision History table and the document links on this page

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